ANALYSE ET EXPLORATIONS DE DONNEES

```
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# Bac2 génie minier / EMI / UOB
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
import numpy as np  # calcul et algebre lineaire
import pandas as pd # gestion de la base de donnees
import matplotlib.pyplot as plt # gestion des graphes  - visualisation
import seaborn as sns # toujours pour la visualisation
```

Chargement de donnees

```
donnees = pd.read_csv('/content/drive/MyDrive/Coffe_sales.csv', sep=',')
 #changer avec la direction de mon téléphone dans Google drive car j'utilise Google colab dans le téléphone
```

Double-cliquez (ou appuyez sur Entrée) pour modifier

```
donnees.head()
```

	hour_of_day	cash_type	money	coffee_name	Time_of_Day	Weekday	Month_name	Weekdaysort	Monthsort	Date	Time
0	10	card	38.7	Latte	Morning	Fri	Mar	5	3	2024-03-01	10:15:50.520000
1	12	card	38.7	Hot Chocolate	Afternoon	Fri	Mar	5	3	2024-03-01	12:19:22.539000
2	12	card	38.7	Hot Chocolate	Afternoon	Fri	Mar	5	3	2024-03-01	12:20:18.089000
3	13	card	28.9	Americano	Afternoon	Fri	Mar	5	3	2024-03-01	13:46:33.006000
4	13	card	38.7	Latte	Afternoon	Fri	Mar	5	3	2024-03-01	13:48:14.626000

```
# Nombre de Lignes(data points) et des colonnes (Features- variables )
print('taille de la de donnees: ',donnees.shape)
print('Nombres de lignes:', donnees.shape[0])
print('Nombres de Colonnes:', donnees.shape[1])
```

```
taille de la de donnees:  (3547, 11)
Nombres de lignes: 3547
Nombres de Colonnes: 11
```

```
# informations sur les types de donnees dans nos colonnes

donnees.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3547 entries, 0 to 3546
Data columns (total 11 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   hour_of_day  3547 non-null   int64  
 1   cash_type    3547 non-null   object 
 2   money        3547 non-null   float64
 3   coffee_name  3547 non-null   object 
 4   Time_of_Day  3547 non-null   object 
 5   Weekday      3547 non-null   object 
 6   Month_name   3547 non-null   object 
 7   Weekdaysort  3547 non-null   int64  
 8   Monthsort    3547 non-null   int64  
 9   Date         3547 non-null   object 
 10  Time         3547 non-null   object 
dtypes: float64(1), int64(3), object(7)
memory usage: 304.9+ KB
```

```
# Verification de Valeurs manquante
donnees.isna().any()
```

	0
hour_of_day	False
cash_type	False
money	False
coffee_name	False
Time_of_Day	False
Weekday	False
Month_name	False
Weekdaysort	False
Monthsort	False
Date	False
Time	False

dtype: bool

```
# Description des valeurs numeriques:

donnees.describe()
```

	hour_of_day	money	Weekdaysort	Monthsort
count	3547.000000	3547.000000	3547.000000	3547.000000
mean	14.185791	31.645216	3.845785	6.453905
std	4.234010	4.877754	1.971501	3.500754
min	6.000000	18.120000	1.000000	1.000000
25%	10.000000	27.920000	2.000000	3.000000
50%	14.000000	32.820000	4.000000	7.000000
75%	18.000000	35.760000	6.000000	10.000000
max	22.000000	38.700000	7.000000	12.000000

```
donnees.describe().style.highlight_max(color='lightgreen',axis=0) # les statistique de vos donnees numerique
```

	hour_of_day	money	Weekdaysort	Monthsort
count	3547.000000	3547.000000	3547.000000	3547.000000
mean	14.185791	31.645216	3.845785	6.453905
std	4.234010	4.877754	1.971501	3.500754
min	6.000000	18.120000	1.000000	1.000000
25%	10.000000	27.920000	2.000000	3.000000
50%	14.000000	32.820000	4.000000	7.000000
75%	18.000000	35.760000	6.000000	10.000000
max	22.000000	38.700000	7.000000	12.000000

```
donnees.describe()[['money']]
```

	money
count	3547.000000
mean	31.645216
std	4.877754
min	18.120000
25%	27.920000
50%	32.820000
75%	35.760000
max	38.700000

```
## types de nos variables

donnees.dtypes
```

	0
hour_of_day	int64
cash_type	object
money	float64
coffee_name	object
Time_of_Day	object
Weekday	object
Month_name	object
Weekdaysort	int64
Monthsort	int64
Date	object
Time	object

dtype: object

```
# Netoyage de donnees :
# 1. Nettoyage de la colonne 'hour_of_day' si nécessaire
donnees['hour_of_day'] = donnees['hour_of_day'].astype(str).str.replace(',', '')
donnees['hour_of_day'] = donnees['hour_of_day'].astype(int)

# 2. Correction pour Month_name
dictionnaire_mois = {
    'Jan': 1, 'Feb': 2, 'Mar': 3, 'Apr': 4, 'May': 5, 'Jun': 6,
    'Jul': 7, 'Aug': 8, 'Sep': 9, 'Oct': 10, 'Nov': 11, 'Dec': 12
}


donnees['Month_name_numeric'] = donnees['Month_name'].map(dictionnaire_mois).astype(float)
```

Double-cliquez (ou appuyez sur Entrée) pour modifier

```
donnees['hour_of_day'].dtype
```

```
dtype('int64')
```

```
# Puisque 'hour_of_day' est déjà un entier (int64),
# on peut directement passer à la suite ou simplement afficher les premières lignes pour vérifier :
donnees['hour_of_day'].head()
```

	hour_of_day
0	10
1	12
2	12
3	13
4	13

dtype: int64

```
# Analyser la distribution des prix payés
print(donnees['money'].describe())
# Compter le nombre de ventes pour chaque type de café
print(donnees['coffee_name'].value_counts())
```

```
count    3547.000000
mean       31.645216
std         4.877754
min        18.120000
25%        27.920000
50%        32.820000
75%        35.760000
max        38.700000
Name: money, dtype: float64
coffee_name
Americano with Milk    809
Latte                  757
Americano              564
Cappuccino             486
Cortado                287
Hot Chocolate          276
Cocoa                  239
Espresso               129
Name: count, dtype: int64
```

```
donnees['money'].head()
donnees['coffee_name'].head()
```

	coffee_name
0	Latte
1	Hot Chocolate
2	Hot Chocolate
3	Americano
4	Latte

dtype: object

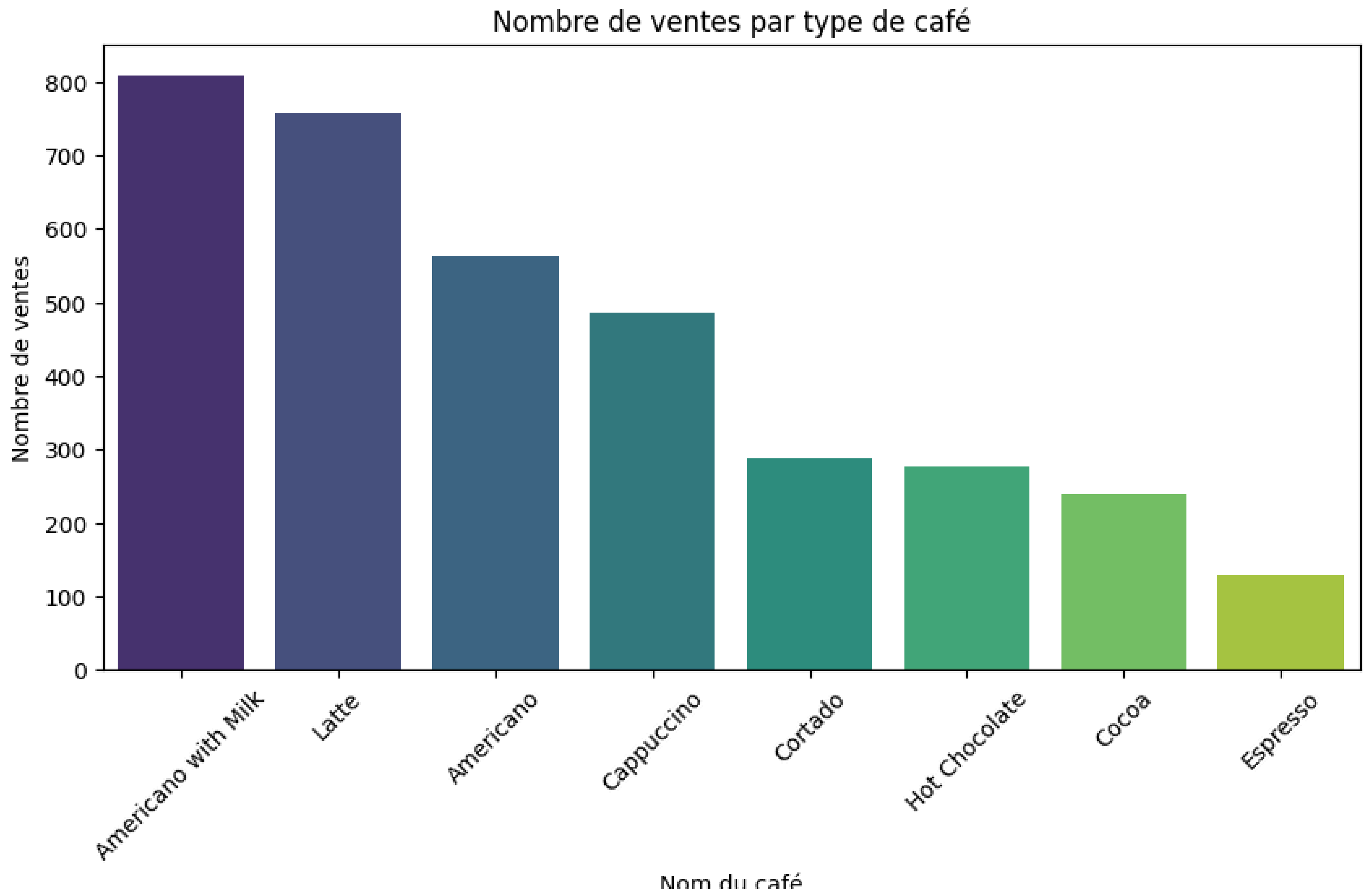
Visualisation

```
plt.figure(figsize=(10, 5))
# On crée un graphique en barres avec le décompte des cafés
sns.countplot(data=donnees, x='coffee_name', order=donnees['coffee_name'].value_counts().index, palette='viridis')
plt.title('Nombre de ventes par type de café')
plt.xlabel('Nom du café')
plt.ylabel('Nombre de ventes')
plt.xticks(rotation=45)  # Incline les noms pour qu'ils soient lisibles
plt.show()
```

```
/tmp/ipykernel_1093/288767161.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(data=donnees, x='coffee_name', order=donnees['coffee_name'].value_counts().index, palette='viridis')
```




```
# Pour les colonnes non-essentielles (n'échoue pas si la colonne n'existe pas)
donnees.drop('Unnamed: 0', axis=1, inplace=True, errors='ignore')
```

```
donnees
```

	hour_of_day	cash_type	money	coffee_name	Time_of_Day	Weekday	Month_name	Weekdaysort	Monthsort	Date	Time	Month_name_numeric
0	10	card	38.70	Latte	Morning	Fri	Mar	5	3	2024-03-01	10:15:50.520000	3.0
1	12	card	38.70	Hot Chocolate	Afternoon	Fri	Mar	5	3	2024-03-01	12:19:22.539000	3.0
2	12	card	38.70	Hot Chocolate	Afternoon	Fri	Mar	5	3	2024-03-01	12:20:18.089000	3.0
3	13	card	28.90	Americano	Afternoon	Fri	Mar	5	3	2024-03-01	13:46:33.006000	3.0
4	13	card	38.70	Latte	Afternoon	Fri	Mar	5	3	2024-03-01	13:48:14.626000	3.0
...	...	...	...	...	...	...	...	...	...	...	...	...
3542	10	card	35.76	Cappuccino	Morning	Sun	Mar	7	3	2025-03-23	10:34:54.894000	3.0
3543	14	card	35.76	Cocoa	Afternoon	Sun	Mar	7	3	2025-03-23	14:43:37.362000	3.0
3544	14	card	35.76	Cocoa	Afternoon	Sun	Mar	7	3	2025-03-23	14:44:16.864000	3.0
3545	15	card	25.96	Americano	Afternoon	Sun	Mar	7	3	2025-03-23	15:47:28.723000	3.0
3546	18	card	35.76	Latte	Night	Sun	Mar	7	3	2025-03-23	18:11:38.635000	3.0

3547 rows × 12 columns

```
## Sauvegarde de donnees pour une manupilation Ulterieure
donnees.to_csv('donnees_emiexo1.csv', index= False)
```

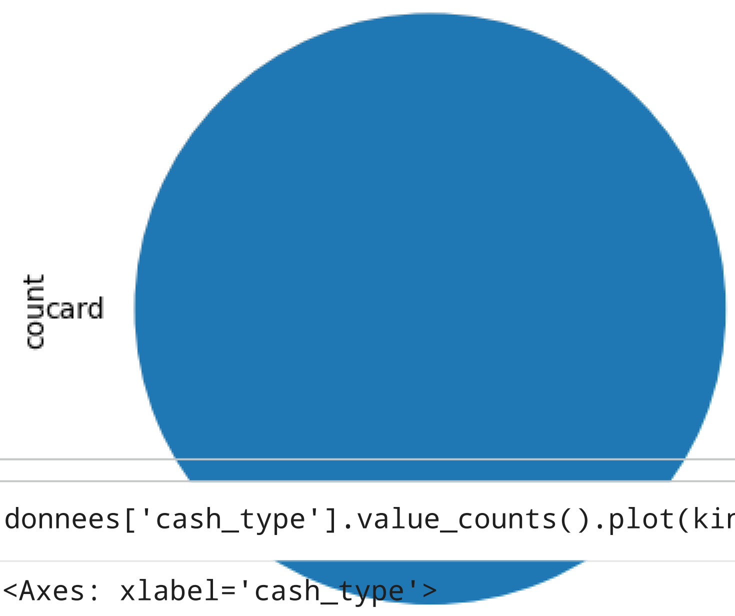
```
# Compte des valeurs dans la colonne des modes de paiement
donnees['cash_type'].value_counts()
```

	count
cash_type	
card	3547

dtype: int64

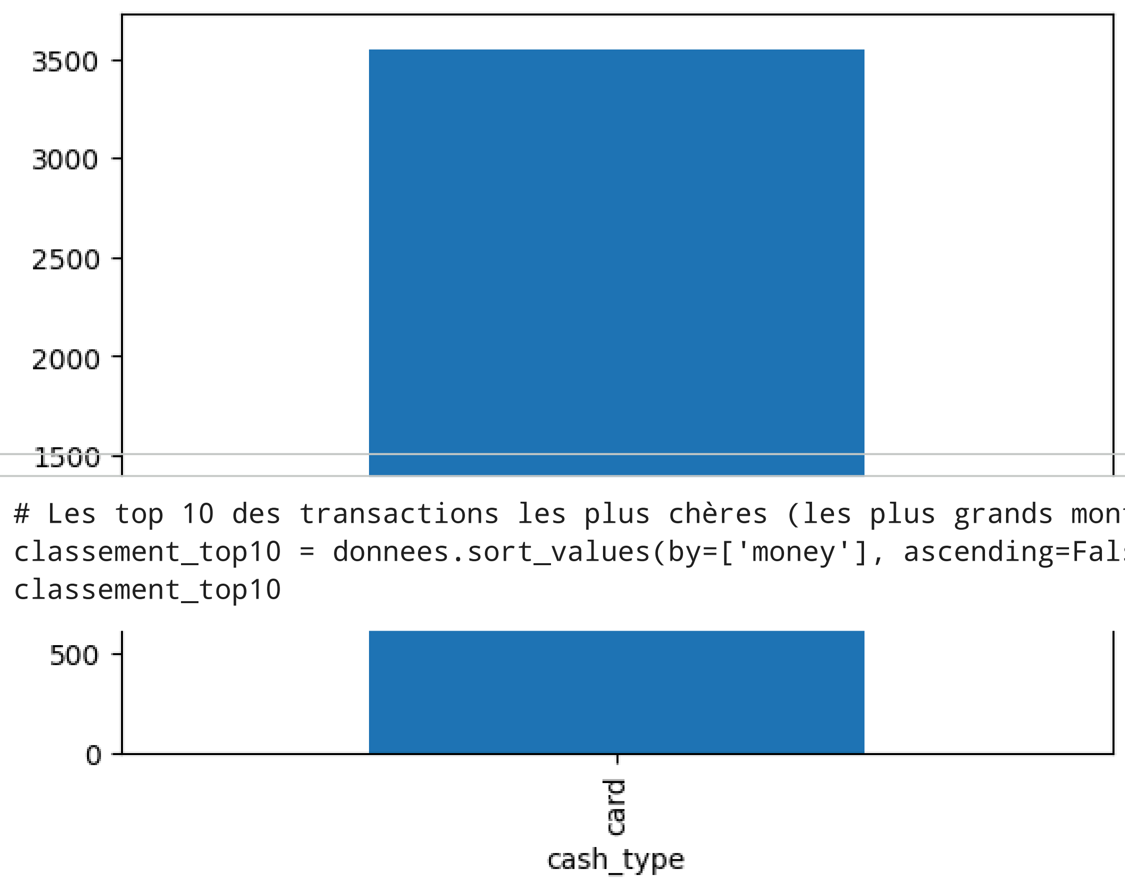
```
donnees['cash_type'].value_counts().plot(kind='pie')
```

<Axes: ylabel='count'>




```
donnees['cash_type'].value_counts().plot(kind='bar')
```

<Axes: xlabel='cash_type'>



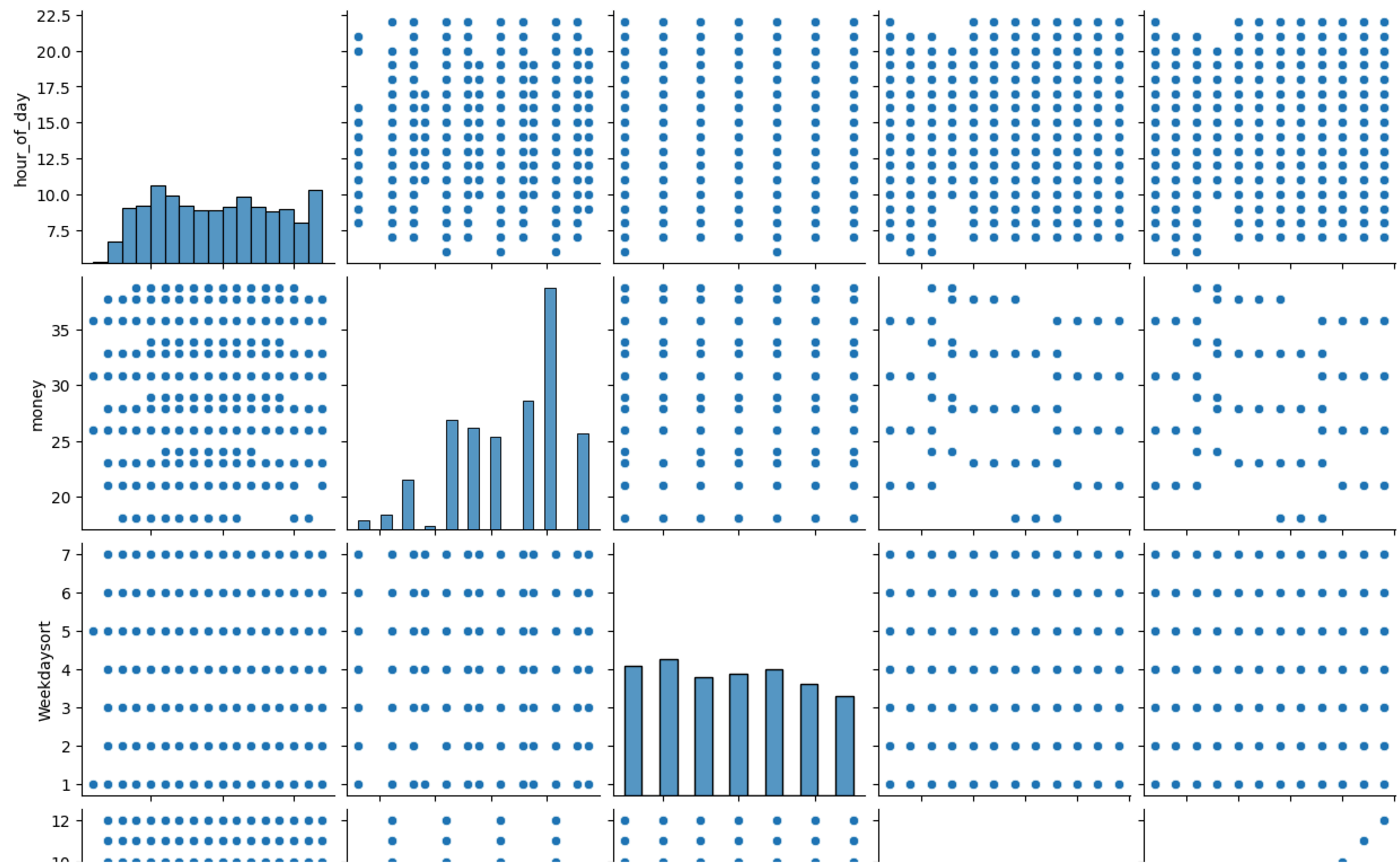

```
# Les top 10 des transactions les plus chères (les plus grands montants de 'money')
classement_top10 = donnees.sort_values(by=['money'], ascending=False).head(10)
classement_top10
```

	hour_of_day	cash_type	money	coffee_name	Time_of_Day	Weekday	Month_name	Weekdaysort	Monthsort	Date	Time	Month_name_numeric
131	10	card	38.7	Latte	Morning	Sat	Mar	6	3	2024-03-23	10:43:59.559000	3.0
133	13	card	38.7	Cappuccino	Afternoon	Sat	Mar	6	3	2024-03-23	13:10:07.661000	3.0
134	13	card	38.7	Cocoa	Afternoon	Sat	Mar	6	3	2024-03-23	13:11:12.230000	3.0
236	12	card	38.7	Cappuccino	Afternoon	Sun	Apr	7	4	2024-04-14	12:24:39.028000	4.0
235	17	card	38.7	Cappuccino	Night	Sat	Apr	6	4	2024-04-13	17:53:19.237000	4.0
234	17	card	38.7	Cappuccino	Night	Sat	Apr	6	4	2024-04-13	17:51:26.026000	4.0
233	16	card	38.7	Latte	Afternoon	Sat	Apr	6	4	2024-04-13	16:19:39.799000	4.0
232	16	card	38.7	Cappuccino	Afternoon	Sat	Apr	6	4	2024-04-13	16:18:03.938000	4.0
259	17	card	38.7	Cappuccino	Night	Wed	Apr	3	4	2024-04-17	17:02:40.687000	4.0
147	10	card	38.7	Latte	Morning	Tue	Mar	2	3	2024-03-26	10:42:30.170000	3.0

```
classement_top10.to_csv('top10_ventes_cafe.csv', index=False)
```

```
## Visualisation de relationes entres toutes les donnees et les distributions

sns.pairplot(donnees)
plt.show()
```




```
donnees_numeriques = donnees.select_dtypes(include=['number'])
```

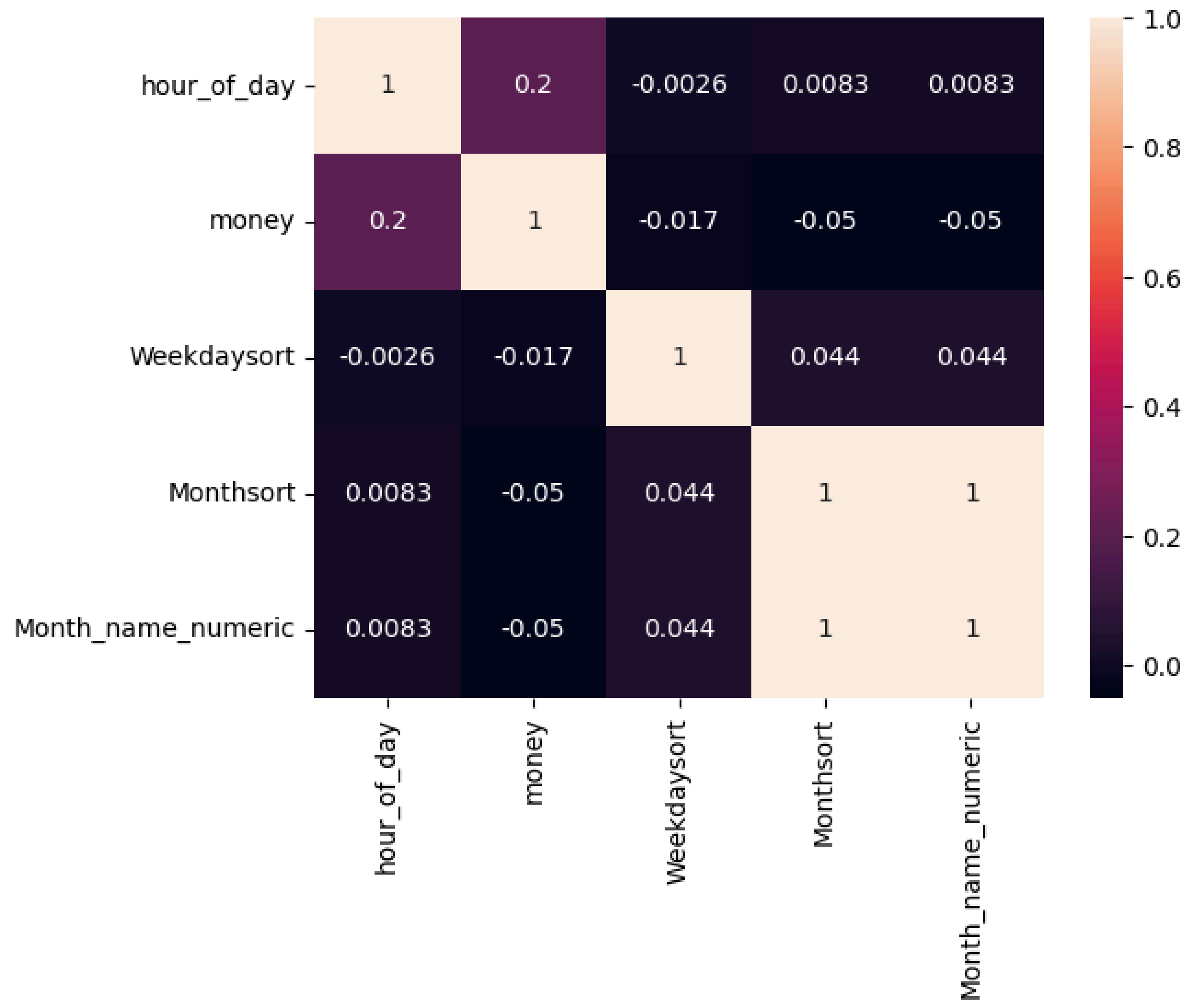
```
donnees_numeriques
```

	hour_of_day	money	Weekdaysort	Monthsort	Month_name_numeric
0	10	38.70	5	3	3.0
1	12	38.70	5	3	3.0
2	12	38.70	5	3	3.0
3	13	28.90	5	3	3.0
4	13	38.70	5	3	3.0
...	...	...	...	...	...
3542	10	35.76	7	3	3.0
3543	14	35.76	7	3	3.0
3544	14	35.76	7	3	3.0
3545	15	25.96	7	3	3.0
3546	18	35.76	7	3	3.0

3547 rows × 5 columns

```
# Correlation entre valeur numerique

sns.heatmap(donnees_numeriques.corr(),  annot = True)
plt.show()
```




```
donnees_numeriques.corr()
```

	hour_of_day	money	Weekdaysort	Monthsort	Month_name_numeric